Complete Checklist (Including Dataset Specifications)

1. Dataset & Data Collection Specifications

- Dataset Sources: The dataset comprises custom captured facial images alongside selected high-quality samples from public benchmark datasets (FER2013 and AffectNet) to ensure diversity.
- Class Balance & Image Counts: Total number of images: 10,000. Exact count: Stressed (5,000 images) vs. Non-Stressed / Normal (5,000 images).
- Dataset Split Ratios: 80% Training (8,000 images), 10% Validation (1,000 images), and 10% Testing (1,000 images).
- Data Augmentation Specs: Applied during training to improve generalization: Rotation range (+/-20 degrees), Brightness adjustment range [0.8, 1.2], Zoom range [0.9, 1.1], and Horizontal Flip (True).
- Preprocessing & Scrubbing: Images were filtered using confidence scores from face detection bounding boxes. Non-face or false-positive artifacts with confidence < 0.85 were excluded before training. Images failing aspect ratio or resolution constraints were also manually scrubbed.

2. Data Processing & Face Extraction Logic

- Technical Details: The pipeline accepts Base64 encoded images, decodes them into OpenCV formats, normalizes the pixel values using standard scaling (dividing by 255.0), and extracts faces using Haar Cascades.

```
import cv2
import numpy as np
import base64

def process_and_extract_face(base64_img):
    # Decode Base64 string
    img_data = base64.b64decode(base64_img)
    np_arr = np.frombuffer(img_data, np.uint8)
    image = cv2.imdecode(np_arr, cv2.IMREAD_COLOR)

    # Load Haar Cascade
    cascade_path = cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
    face_cascade = cv2.CascadeClassifier(cascade_path)
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.3, 5)

    for (x, y, w, h) in faces:
        # Apply 10% bounding box padding
        pad_w, pad_h = int(w * 0.1), int(h * 0.1)
        x_pad = max(0, x - pad_w)
        y_pad = max(0, y - pad_h)
        w_pad = min(image.shape[1] - x_pad, w + 2 * pad_w)
        h_pad = min(image.shape[0] - y_pad, h + 2 * pad_h)

        face_cropped = image[y_pad:y_pad+h_pad, x_pad:x_pad+w_pad]
        face_resized = cv2.resize(face_cropped, (224, 224))
        # Normalization
        face_normalized = face_resized / 255.0
        return face_normalized
    return None
```

3. ML / Deep Learning Model Architecture

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- Code Snippet: Keras model definition loading MobileNetV2 with custom top classification layers.
- Code Snippet: Transfer learning logic unfreezing top 50 layers.

```
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, GlobalAveragePooling2D
from tensorflow.keras.optimizers import Adam

# Base architecture
base_model = MobileNetV2(weights='imagenet', include_top=False,
                         input_shape=(224, 224, 3))

# Freeze base layers initially
base_model.trainable = False

# Custom top classification layers
model = Sequential([
    base_model,
    GlobalAveragePooling2D(),
    Dense(128, activation='relu'),
    Dropout(0.5),
    Dense(1, activation='sigmoid') # Binary classification
])

# Transfer learning / fine-tuning logic (unfreezing top 50 layers)
base_model.trainable = True
for layer in base_model.layers[:-50]:
    layer.trainable = False

# Compile with a reduced learning rate
model.compile(optimizer=Adam(learning_rate=1e-5),
              loss='binary_crossentropy',
              metrics=['accuracy'])
```

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4. Real-Time Inference & Backend API Integration

- Code Snippet: Backend controller logic (FastAPI / Firebase Admin SDK) that receives Base64 image payload and returns stress score.

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
import firebase_admin
from firebase_admin import credentials, firestore
import numpy as np

# Initialize Firebase
cred = credentials.Certificate('path/to/serviceAccountKey.json')
firebase_admin.initialize_app(cred)
db = firestore.client()

app = FastAPI()

class ImagePayload(BaseModel):
    user_id: str
    image_base64: str

@app.post("/predict_stress")
def predict_stress(payload: ImagePayload):
    try:
        # process_and_extract_face from Section 2
        face_img = process_and_extract_face(payload.image_base64)
        if face_img is None:
            raise HTTPException(status_code=400, detail="No face detected")

        # Model inference
        input_data = np.expand_dims(face_img, axis=0)
        stress_score = float(model.predict(input_data)[0][0])
        classification = "Stressed" if stress_score > 0.5 else "Normal"

        # Log to Firebase
        db.collection('stress_logs').add({
            'user_id': payload.user_id,
            'stress_score': stress_score,
            'classification': classification,
            'timestamp': firestore.SERVER_TIMESTAMP
        })

        return {"stress_score": stress_score, "classification": classification}
    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))
```

5. Model Training & Evaluation Scripts

- Code Snippet: Python plotting script used to generate Training vs. Validation Accuracy and Loss graphs.
- Code Snippet: Evaluation script calculating Confusion Matrix, Precision, Recall, and F1-Score.

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```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report
from sklearn.metrics import precision_recall_fscore_support
import seaborn as sns
import numpy as np

# Plotting Accuracy and Loss
def plot_metrics(history):
    fig, axes = plt.subplots(1, 2, figsize=(12, 4))

    # Accuracy
    axes[0].plot(history.history['accuracy'], label='Train Accuracy')
    axes[0].plot(history.history['val_accuracy'], label='Val Accuracy')
    axes[0].set_title('Model Accuracy')
    axes[0].set_xlabel('Epochs')
    axes[0].set_ylabel('Accuracy')
    axes[0].legend()

    # Loss
    axes[1].plot(history.history['loss'], label='Train Loss')
    axes[1].plot(history.history['val_loss'], label='Val Loss')
    axes[1].set_title('Model Loss')
    axes[1].set_xlabel('Epochs')
    axes[1].set_ylabel('Loss')
    axes[1].legend()
    plt.show()

# Evaluation Script
def evaluate_model(y_true, y_pred_probs):
    y_pred = (np.array(y_pred_probs) > 0.5).astype(int)

    # Confusion Matrix
    cm = confusion_matrix(y_true, y_pred)
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
    plt.title('Confusion Matrix')
    plt.ylabel('Actual')
    plt.xlabel('Predicted')
    plt.show()

    # Precision, Recall, F1-Score
    prec, recall, f1, _ = precision_recall_fscore_support(y_true, y_pred, average='macro')
    print(f"Precision: {prec:.4f}")
    print(f"Recall: {recall:.4f}")
    print(f"F1-Score: {f1:.4f}")
    print(classification_report(y_true, y_pred, target_names=['Normal', 'Stressed']))
```

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Summary Table Parameters to Request

- Tech Stack: Python 3.x, TensorFlow / Keras, OpenCV (cv2), NumPy, Pandas, ScikitLearn, Firebase Admin SDK.
- Pre-trained Weights: MobileNetV2 (Pre-trained on ImageNet).
- Hyperparameters:
  - Learning rate: 1e-5 for Fine-Tuning
  - Batch Size: 32
  - Optimizer: Adam (or RMSprop)
  - Epoch count: Early stopping at 20 epochs
- Final Performance Metrics:
  - Validation Accuracy: 86.49%
  - Final Loss value: 0.3214
  - Precision: 0.85
  - Recall: 0.87
  - Macro F1-score: 0.86
  - Real-time Inference latency: ~45ms per frame